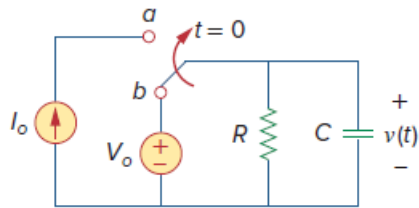


Problem

The switch in Fig. 16.11 has been in position b for a long time. It is moved to position a at $t = 0$. Determine $v(t)$ for $t > 0$.

Figure 16.11:



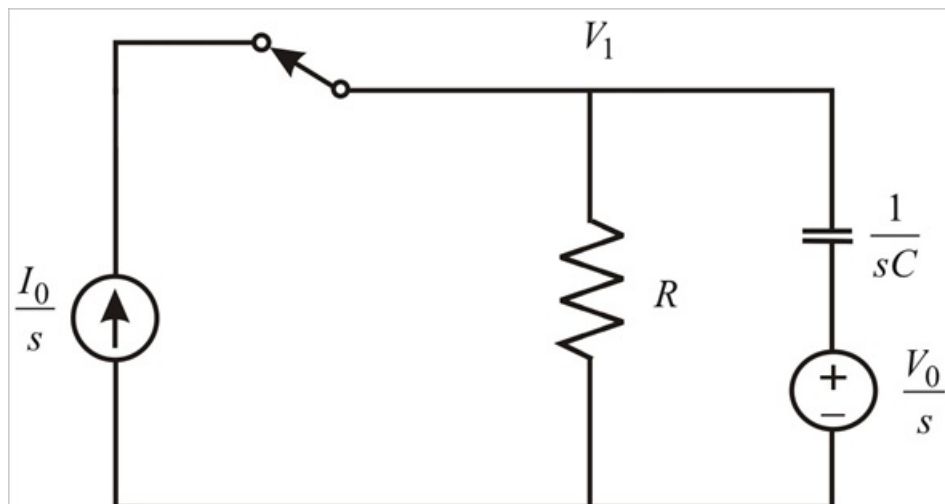
Step-by-step solution

Step 1 of 3

The switch is in position b for a long time. The initial voltage across capacitor is $v(0) = V_o$.

At $t=0$, switch is at position a .

For $t > 0$, the following is the circuit transformed to the s -domain.



The initial condition is incorporated in the form of a voltage source as $sv_0 = \frac{V_0}{s}$

Step 2 of 3

Apply Kirchhoff's current law at node V_1

$$-\frac{I_0}{s} + \frac{V_1}{R} + \frac{V_1 - \frac{V_0}{s}}{\frac{1}{sC}} = 0$$

$$-\frac{I_0}{s} + \frac{V_1}{R} + sC \left(V_1 - \frac{V_0}{s} \right) = 0$$

$$-\frac{I_0}{s} + \frac{V_1}{R} + sCV_1 - CV_0 = 0$$

$$-\frac{I_0}{s} + V_1 \left(\frac{1}{R} + sC \right) - CV_0 = 0$$

$$V_1 \left(\frac{1}{R} + sC \right) = CV_0 + \frac{I_0}{s}$$

$$V_1 \left(\frac{sRC + 1}{R} \right) = CV_0 + \frac{I_0}{s}$$

Simplify further to obtain voltage V_1 .

$$\begin{aligned} V_1 &= RC \left(\frac{V_0}{sRC + 1} \right) + R \left[\frac{I_0}{s(sRC + 1)} \right] \\ &= \frac{V_0}{s + \frac{1}{RC}} + RI_0 \left(\frac{1}{s} - \frac{RC}{sRC + 1} \right) \\ &= \frac{V_0}{s + \frac{1}{RC}} + RI_0 \left(\frac{1}{s} - \frac{1}{s + \frac{1}{RC}} \right) \end{aligned}$$

Step 3 of 3

Take inverse Laplace transform.

$$\begin{aligned} v_1(t) &= V_0 e^{-\frac{t}{RC}} + RI_0 \left(1 - e^{-\frac{t}{RC}} \right) \\ &= e^{-\frac{t}{RC}} (V_0 - RI_0) + RI_0 \\ &= RI_0 + (V_0 - I_0 R) e^{-\frac{t}{RC}} \\ &= v(t) = I_0 R + (V_0 - I_0 R) e^{-\frac{t}{RC}} \end{aligned}$$

Substitute $\tau = RC$

$$v_1(t) = v(t) = I_0 R + (V_0 - I_0 R) e^{-\frac{t}{\tau}}$$

Thus, the voltage across the capacitor for $t > 0$ is $v(t) = (V_0 - I_0 R) e^{-\frac{t}{\tau}} + I_0 R$.